Separations without Unemployment: the Escape Margin across Local Labor Markets

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Abstract

Unemployment rates vary largely across local labor markets. Recent studies have shown that these differences are mainly driven by differences in the rate by which workers transition from employment to unemployment (EU) rather than variation in the job-finding rate among the unemployed. This paper revisits this decomposition and studies an additional channel by which job-finding can affect local unemployment rates: when a worker receives a separation shock, the worker can search for a new job while still employed, turning a potential transition into unemployment (EU) into a direct employer-to-employer move (EE). I call this the escape margin of separations.

I develop a structural continuous-time job search model with a notice state, estimated on matched Danish employer–employee data covering 148 labor market segments (4 regions $\times$ 37 occupations). The model recovers latent separation rates and escape probabilities that are not separately observed in the data. I use the estimated model to decompose the cross-sectional variation in unemployment rates into separation rates and job-finding rates, including the escape margin. The results show that separations remain the primary driver of unemployment variation, but the additional channel of the escape margin significantly increase the importance of job-finding.

1 Introduction

Many of the richest countries in the world have experienced increasing regional disparities[1] over the last two decades. Growth has been concentrated in a few metropolitan areas, while other regions have stagnated. This affects employment opportunities directly, but also has indirect effects through worsening local public finances and wealth effects through the housing market. The consequences

*Job Market paper

[1]Measured by regional GDP per capita. OECD (2023).

are both economic and political: it stands as an important explanation for the increasing political polarization between rural and urban areas, where workers in lagging regions express their discontent by voting for anti-establishment parties (Autor et al., 2020).

Unsurprisingly, this has attracted the attention of policymakers, who have implemented a range of policies to create growth outside metropolitan areas. In the US, a lot of ressources are allocated to place-based policies (Moretti, 2011), while the EU commision has recently launch the Right to Stay strategy with the ambition to create more equal opportunities across regions in European countries.

Unemployment rates provide a clear example of the regional disparities as it has been documentated across many countries that unemployment rates vary largely and persistently across locations (Kline and Moretti, 2013; Kuhn et al., 2026). Recent research attributed this variation to the fact that the rate at which workers transition from employment to unemployment (EU) is highly variable across regions, while the rate at which unemployed workers find jobs (UE) vary less with unemployment rates (Bilal, 2023; Kuhn et al., 2026). The variance decomposition in Kuhn et al. (2026) shows that two thirds of the variation in unemployment rates is related to regional differences in EU transitions, while one third of the unemployment is ascriptable to the rate at which unemployed workers find jobs. This finding is surprising when set against its time-series counterpart where studies have shown that variation in unemployment rates over the business cycle is mainly driven by fluctuations in the job-finding rate among the unemployed (Shimer, 2012).

In this description of labor flows, job finding can affect unemployment only through the rate at which unemployed workers find jobs. In this paper, I introduce an additional channel through which job finding affects unemployment: job finding among employed workers can determine whether a separation shock results in unemployment in the first place.

The standard decomposition of unemployment into EU and UE flows implicitly treats separations leading to unemployment as a primitive transition. Instead, I distinguish between the underlying separation shock and the observed transition into unemployment. When a worker receives a separation shock, she may find a new job before the current employment relationship ends. A separation shock can therefore result either in an EE transition or an EU transition. The observed EU rate consequently depends not only on the rate of separation shocks, but also on workers' probability of finding jobs while still employed. I call this the escape margin of separations.

I argue that this is a realistic and empirically relevant mechanism when studying cross-sectional differences in unemployment rates. One reason is that it has a real world counterpart in the form of notice periods. In many countries, workers are entitled to a notice period before a layoff, during which they continue to receive their wage and can search for new employment. The existence of notice periods allows workers to search for new employment while still employed after having received a separation shock. This is universal among OECD countries: in all of them, firms are required to inform workers in advance of a layoff

(Cederlöf et al., 2025). In Denmark, the notice window is also long enough to matter quantitatively. As I document, the majority of unemployed unemployed workers find a job within four months and workers are on average entitled to a notice period. If notified workers search as efficiently as unemployed workers, this means that a majority of workers receiving a separation shock have a substantial window of time in which to line up a new employer before entering unemployment. This might even be a conervative assumption since (Cederlöf et al., 2025) finds that noticed workers search even more efficiently than unemployed workers. The escape margin is therefore likely to be quantitatively important in the Danish context, and in other countries with mandatory notice periods.

There are good reasons to expect this channel is relevant for the aggregate level of unemployment but also for the cross-sectional variation in unemployment rates. Previous research has shown that workers in denser labor markets are more likely to make direct employer-to-employer transitions (Schmutz and Sidibé, 2019). If dense labor markets make it easier for workers facing an impending separation to find another job before becoming unemployed, part of the variation in observed separation rates are due to better job-finding opportunities. This is also consistent with the findings in Moretti and Yi (2024) that workers in denser labor markets have better employment outcomes after experiencing a seperation shock through a mass layoff.

I provide empirical evidence that this mechanism is relevant in the Danish labor market. Across labor market segments, I show that the observed EU transition rate is positively correlated with EE transition rates, consistent with a model in which workers receive separation shocks at different rates across segments and some workers escape unemployment by finding a new job before the separation results in unemployment.

I provide additional evidence for the mechanism by studying wage changes associated with EE transitions. Workers in high-unemployment segments are more likely to accept a wage cut when changing jobs. Such downward wage mobility is difficult to reconcile with a standard job-ladder model, in which employed workers only accept offers that improve upon their current job. For this reason, the job-ladder literature has introduced reallocation shocks and "Godfather" shocks that can induce workers to move to lower-paying jobs [REFERENCES].

The mechanism proposed here differs from these approaches in two important respects. First, receiving a separation shock does not force the worker to accept the next offer. A notified worker remains employed during the notice period and can reject an offer and continue searching for a better one, but does so knowing that unemployment is imminent if no acceptable job is found. The separation shock therefore lowers the worker's reservation wage while preserving a non-trivial acceptance decision. Second, the probability of escaping unemployment through an EE transition depends on the labor market segment in which the worker is currently employed. Differences in local job-finding opportunities therefore determine whether otherwise similar separation shocks result in EE transitions or entries into unemployment.

These correlations suggest that the escape margin plays an important role in shaping unemployment rates in the Danish labor market. They cannot, however, reveal its quantitative importance. The observed correlations are equilibrium outcomes, and EE transitions cannot be separated into those that follow a separation shock and those that would have occurred otherwise.

To quantify the relative contributions of separation rates, the escape margin, and job finding among unemployed workers to differences in unemployment rates, I therefore develop and estimate a structural model of worker flows.

In this model, workers search both on-the-job and off-the-job. They search across labor market segments defined by pairing regions and occupations. This segmentation is motivated by a descriptive analysis which shows that worker flows follow gravity patterns in both the geographical and occupational dimension. The model features a notice state: when a worker receives a separation shock, she continues to earn her wage and search for new opportunities for a statutorily determined notice period before the match ends and she enters unemployment. During this window, she can escape into a new job. The model thus endogenizes whether a separation shock results in an EE transition or an EU transition.

The estimated model allows me to recover the latent objects that cannot be observed directly in the data: the incidence of separation shocks and the probability that a worker finds another job before entering unemployment. I can then use counterfactuals to quantify how much of the cross-sectional variation in unemployment is attributable to differences in these escape probabilities, and hence to job-finding opportunities that are hidden from conventional decompositions based on observed EU and UE transition rates.

Estimating the model on Danish matched employer–employee data, I find that the escape margin reveals an additional channel through which job-finding rates affect unemployment. Job-finding operates through two margins—the standard UE channel and the escape margin from the notice state—which together account for 29% of the explained cross-segment variation in unemployment rates, compared to 22% for the UE channel alone. This suggests that part of what appears to be a job-loss problem across space is, in fact, a job-finding problem—one that is hidden inside the measured EU transition rate.

These findings have direct implications for the design of place-based policies. If spatial unemployment differences are partly driven by low escape probabilities in thin labor markets, then an increase job finding opportunities will affect unemployment rates both through the escape margin and through the standard channel of job finding among the unemployed.

Related Literature This paper relates to four strands of literature. First, it contributes to the recent literature studying the determinants of unemployment differences across labor market segments (Bilal, 2023; Kuhn et al., 2026). This literature finds that cross-sectional differences in unemployment are primarily associated with differences in employment-to-unemployment (EU) transition rates rather than unemployment-to-employment (UE) transition rates. I show that

this interpretation can overlook an important role for job finding. When workers facing separation can find a new job before entering unemployment, differences in job-finding opportunities affect unemployment not only through the outflow from unemployment, but also by determining whether a separation results in unemployment in the first place. This escape margin has previously been studied in the context of aggregate unemployment fluctuations over the business cycle (Simmons, 2023). I instead study its role in explaining cross-sectional differences in unemployment across labor market segments.

Second, the paper relates to the literature on spatial job search and worker mobility (Kennan and Walker, 2011; Schmutz and Sidibé, 2019). The search model builds particularly on Schmutz and Sidibé (2019), in which workers search across spatially separated labor markets and face geographic search frictions. I extend this framework by allowing workers to search simultaneously across locations and occupations. This makes it possible to distinguish the geographic and occupational margins through which workers respond to differences in job opportunities across labor market segments.

Third, the occupational dimension connects the paper to the literature on occupational mobility and occupation-specific human capital (Kambourov and Manovskii, 2009; Bocquet, 2024). As in Bocquet (2024), occupational mobility in the data exhibits a gravity structure: workers are more likely to move between occupations that are similar in their skill requirements. I incorporate this structure into the search model by allowing search frictions to depend on skill-based occupational distance.

Finally, the paper relates to the literature on the estimation of high-dimensional dynamic discrete-choice models. In particular, I build on the approach of Arcidiacono et al. (2023) to make estimation tractable in a setting with a large number of labor market states and transition parameters. This allows the model to retain rich heterogeneity in workers' search opportunities across locations and occupations while remaining empirically estimable.

This paper proceeeds by presenting the data and motivating empirical facts (2), which I use to motivate the structural job-search (3). I then present the estimation strategy (??) and the results (??) before concluding (??).

2 Data and Empirical Evidence

The empirical analysis uses administrative matched employer–employee data from Denmark covering every employment spell of wage earners[2] from 2008 to 2019. These data contain exact start and end dates, which means that labor market transitions can be measured at the daily level. The high frequency of the spell data mitigates concerns about time-aggregation (Shimer, 2012) when measuring labor market transitions. This allows me to calculate hazard rates at the daily level as the number of events divided by the total number of days workers were at risk of making a particular transition. The data also include job-specific information such as wage, working hours, occupation, and workplace

[2] Beskæftigelse for lønmodtagere, BFL

location. I use other registers to obtain demographic information about workers [3].

Table 1 summarizes the sample selection. The initial sample contains approximately 4 million unique workers and 16 million employment spells. I restrict the sample to workers who have completed their education and are between 20 and 69 years old. I further require the state variables used in the model—location, occupation, and wage—to be observed. These restrictions result in a final sample of approximately 3 million workers and 9 million employment spells.

Table 1: Sample selection

	Workers	Spells	Worker share	Spell share
All	4,004,254	15,939,484	1.00	1.00
Completed education	3,636,269	12,654,144	0.91	0.79
Aged 20–60	3,175,923	11,135,702	0.79	0.70
Non-missing location	3,103,707	10,610,592	0.78	0.67
Non-missing occupation	2,817,975	9,010,389	0.70	0.57
Non-missing wage	2,817,970	9,010,113	0.70	0.57

The employment data track workers only while they are employed. I therefore supplement these data with the DREAM database, which contains information on individuals receiving social benefits. I use this information to determine whether a gap between two employment spells reflects a period of unemployment or is part of a direct job-to-job transition.

Specifically, I classify a transition between two employment spells as job-to-job (EE) if the worker does not receive unemployment benefits at any point between the two spells. If the worker receives unemployment benefits during the intervening period, I instead construct an unemployment spell between the two employment spells. The resulting transition sequence is then an employment-to-unemployment (EU) transition followed by an unemployment-to-employment (UE) transition into the subsequent job. If no unemployment benefits are received between the employment spells, I treat the move as a direct EE transition.

Measuring unemployment Unemployment is measured from DREAM benefit codes: a gap between two employment spells counts as an unemployment spell only if the worker receives unemployment benefits during the gap. Unemployment is thus measured by benefit receipt rather than by the standard definition of being without a job and actively searching. This misses a group of unemployed workers who do not receive transfers—for example workers who are not entitled to benefits, have exhausted their entitlement, or choose not to

[3] I obtain workers' educational background from UDDA (Uddannelser), work experience from IDA (IDA persondata), and demographics such as age, marital status from BEF (Befolkningen)

claim. For these workers, the move into the next job is classified as EE even though they have been without a job and searching at the same time.

I argue that this is a minimal problem for two reasons. First, the group of unemployed non-recipients is small in Denmark: benefit recipiency among the unemployed is high, and XX% of workers who separate from a job receive unemployment benefits within YY months of the separation. Second, the misclassification is bounded in scope: it only reclassifies transitions for workers who move directly between jobs without receiving benefits, and these spells are short—workers who do not claim benefits find new jobs quickly, with gaps of less than one month (Figure 1)—so relatively few transitions are affected.

Figure 1 show the distribution of gaps between employment spells for EE and EUE transitions. The left panel shows that EE transitions (without UI benefits) typically involves less than 1 months of gap between employment spells, while the gap is longer for transitions that involve a period of unemployment (EUE) with an average of 6 months.

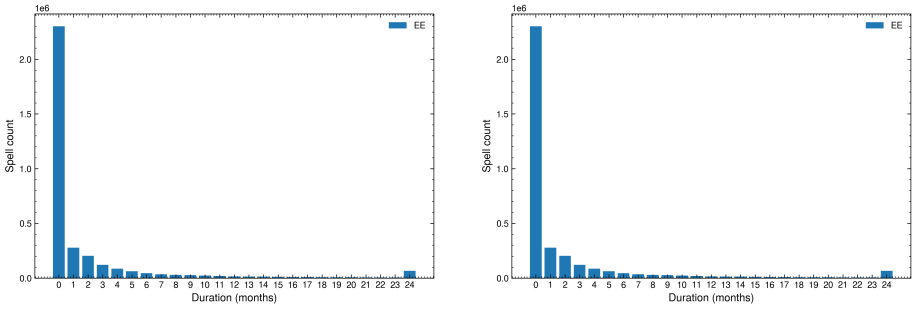


Figure 1: Gap (months) between employment spells for EE and EUE transitions

Spatial and Occupational Segmentation As in much of the literature, I define labor market segments by interacting geographic areas with job types. Previous studies have used different dimensions to characterize jobs, including firms and industries; in this paper, I use occupations. Occupations are defined using the DISCO-08 classification, the Danish version of ISCO-08. I use the two-digit level, which contains 39 occupational groups. I exclude group 95, which is very rare in the Danish labor market, and merge groups 61 and 62, leaving 37 occupational groups. This level of aggregation balances the need for sufficiently large groups against the objective of capturing meaningful differences in skill requirements and career paths across occupations.

The geographic segmentation involves a similar trade-off. I aggregate the 10 Danish provinces into four regions: Copenhagen, Zealand, Jutland, and Funen, shown in Figure 2. The aggregation balances three considerations. First, regions should be sufficiently large to approximate distinct local labor markets, with relatively limited commuting across regional boundaries. If regions were defined too narrowly, workers could readily access jobs in neighboring regions without

relocating, making the boundaries less meaningful for the set of employment opportunities available to them. Second, regions should be sufficiently disaggregated to capture meaningful geographic heterogeneity in the composition of jobs and workers. Finally, because the computational complexity of estimating the model increases with the number of labor market segments, a relatively parsimonious geographic classification is desirable.

Combining the four regions with the 37 occupational groups yields 148 labor market segments.

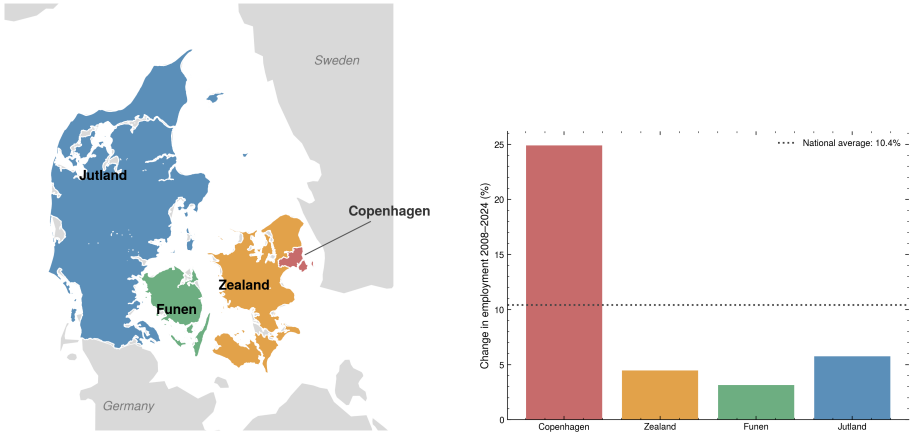


Figure 2: Spatial segmentation of the labor market into four regions. Employment by workplace location. Source: Statistics Denmark, RAS301.

For the rest of the analysis, I define local labor markets as the 148 segments obtained by pairing the four locations with the 37 occupational groups.

Central for the analysis is that these labor market segmetns differ substantially in their unemployment rates, and that these differences are persistent over time. The next section documents this pattern and the associated transition rates (Figure 3)

2.1 Motivating Facts

I begin by documenting how unemployment and worker transition rates vary across labor market segments. Figure 4 plots segment-level unemployment rates against the three transition rates: employment-to-unemployment (EU), unemployment-to-employment (UE), and employment-to-employment (EE).

The first two panels reproduce a pattern previously documented across geographic labor markets (Bilal, 2023; Kuhn et al., 2026). Unemployment is strongly positively correlated with the EU transition rate, whereas its relationship with the UE transition rate is considerably weaker. Viewed through the standard two-state accounting framework, this pattern suggests that differences in unemployment across labor markets are primarily associated with differences

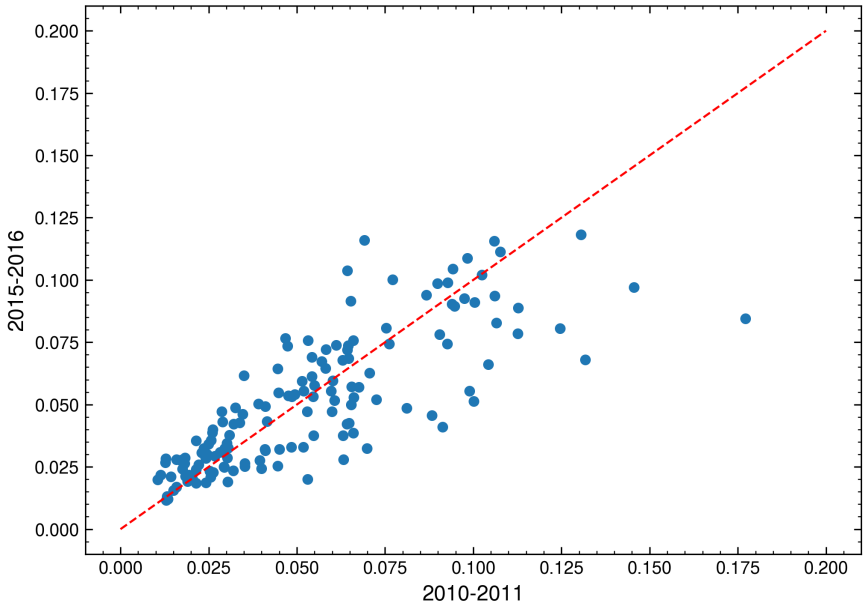


Figure 3: Persistence of segment unemployment rates across the cross-section.

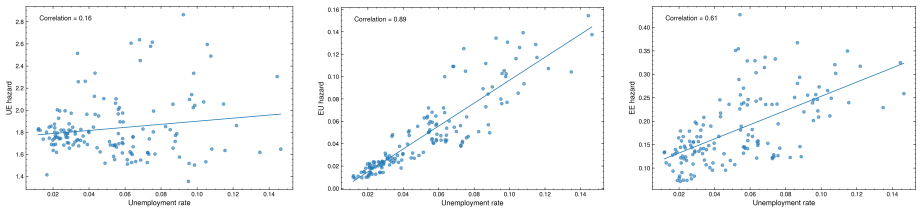


Figure 4: Segment transition hazards against segment unemployment rates.

in job separations rather than job finding. The third panel, however, shows an additional empirical fact: EE transition rates are also positively correlated with unemployment. Thus, high-unemployment segments are characterized not only by more transitions from employment into unemployment, but also by more direct transitions between employers.

A simple way to formalize this argument is to decompose the observed transition hazards as

$$\begin{aligned} EU_s &= \delta_s(1 - e_s), \\ EE_s &= \delta_s e_s + p_s^e, \\ UE_s &= f_s, \end{aligned} \tag{1}$$

where δ_s denotes the rate sat which employed workers in segment s receive separation shocks, e_s is the probability of finding a new job following a separation shock but before entering unemployment, p_s^e captures job-to-job transitions unrelated to an impending separation, and f_s is the job-finding rate from unemployment.

This decomposition provides a natural interpretation of the positive correlation between EU and EE transitions observed in the data. Both transition rates partly reflect the underlying separation rate, δ_s: more separation shocks generate more entries into unemployment, but also more opportunities for workers to escape unemployment through a direct transition to another employer. The observed EU rate is therefore not a primitive separation rate. Rather, it is an equilibrium outcome determined jointly by the arrival of separation shocks and workers' ability to find another job before the employment relationship ends.

This distinction also changes how the relationship between EE transitions and unemployment should be interpreted. Unconditionally, segments with high separation rates may exhibit both high unemployment and high EE transition rates, generating a positive relationship between the two. Conditional on separation risk, however, a higher probability of escaping a separation through a direct job-to-job transition reduces the probability that a separation results in unemployment, lowering the unemployment rate. The model therefore suggests that the conditional relationship between EE transitions and unemployment can have the opposite sign from their unconditional relationship.

To examine this implication in the data, I regress segment-level unemployment rates on the observed transition hazards. The results are reported in Table 2. Column (1) reproduces the unconditional pattern shown in Figure 4: EE transition rates are positively associated with unemployment across segments. Column (2) controls for the EU transition rate. Once differences in EU rates are taken into account, the coefficient on EE becomes negative. Column (3) additionally controls for UE rates, and the negative relationship between EE and unemployment remains. This reversal is consistent with the interpretation suggested by the decomposition: the unconditional positive relationship between EE and unemployment reflects, at least in part, common variation in latent separation risk, whereas conditional on this component, more EE transitions are associated with lower unemployment because it implies that more workers escape before transitioning into unemployment.

Table 2: Segment-level unemployment rates regressed on observed transition hazards.

	(1)	(2)	(3)	(4)
hazard_ee	0.244***	-0.134***	-0.152***	-0.201***
	(0.032)	(0.024)	(0.016)	(0.043)
hazard_eu		1.016***	1.172***	1.163***
		(0.065)	(0.045)	(0.146)
hazard_ue			-0.036***	-0.019*
			(0.003)	(0.010)
Constant	0.009*	0.028***	0.090***	
	(0.005)	(0.003)	(0.006)	
Constant				0.052***
				(0.016)
R-squared	0.369	0.839	0.921	0.976
R-squared Adj.	0.365	0.837	0.919	0.967
Observations	148	148	148	148

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

The first column confirm the firnding from the plot (figure 4) that EE transitions are unconditionally positively correlated with unemployment. This picture immidiately change when looking at the relation controlling for EU rates, which makes the relationship between I run a regression of the segment-level unemployment rate on the three transition rates. The results are reported in Table 2. Column (1) shows that the EU transition rate is strongly positively correlated with unemployment, while the UE transition rate is not. Column (2) adds the EE transition rate to the regression, and shows that it is also positively correlated with unemployment. Column (3) includes all three transition rates, and shows that the positive correlation between EE and unemployment remains even after controlling for EU and UE.

Not all EE transitions, however, need to occur in response to a separation shock. Workers may also move between employers through ordinary on-the-job search. Allowing for such transitions, the observed EE rate can be written as

$$EE_s = \delta_s p_s^S + p_s^E, \tag{2}$$

where p_s^E captures other job-to-job transitions. The positive correlation between EE and unemployment therefore does not by itself identify the post-separation job-finding margin. It does, however, motivate distinguishing the underlying separation process from the observed EU transition and quantifying the extent to which job finding following a separation prevents workers from entering unemployment. This pattern motivates a distinction between an underlying separation shock and the observed transition into unemployment. Suppose that

workers receive separation shocks at rate δ_s. Following such a shock, a worker may either enter unemployment or find another job sufficiently quickly to transition directly to another employer. If p_s^S denotes the probability of the latter outcome, observed EU transitions are given by

$$EU_s = \delta_s(1 - p_s^S).$$

Some EE transitions may therefore reflect successful job finding around a separation rather than ordinary on-the-job mobility.

The positive correlation between EE and unemployment does not by itself identify this mechanism. In particular, EE transitions may also arise from ordinary on-the-job search, and variation in the underlying separation rate can generate a positive correlation between both EU and EE transitions. Rather, the evidence highlights a limitation of interpreting the observed EU rate as a primitive separation rate. Whether a separation shock results in unemployment depends on workers' ability to find another job before entering unemployment.

The quantitative analysis therefore distinguishes between the arrival of separation shocks and the worker's subsequent employment outcome. This allows me to assess a margin that is absent from the standard EU–UE decomposition: job finding can affect unemployment not only by moving unemployed workers into employment, but also by preventing separating workers from entering unemployment in the first place.

Before turning to the structural model, I present descriptive evidence of worker mobility across segments, which motivates the model's segmentation and job search structure.

2.2 Worker Mobility

Table 3 summarizes worker flows across segments. Each transition is classified along two dimensions: whether it changes the worker's location and occupation, and the associated labor market transition which can be either job-to-job transitions (EE) or into unemmploment (UE). The last category is workers that stay within a spell, but where I record their starting location and occupations and the corresponding variaables in the last month. Transitions into unemploment is not presented in the table because they – per defintion – does not envolve a change of segment, because I segments are solely determined by the workplace I assume that unemployed workers stay within the segment of their most recent employment spell. The table therefore only shows transitions that involve a change of segment, which are either EE transitions or the first transition into unemployment (EU).

The previous sections showed that workers frequently move between segment, and especially when transitioning out of unemployment. In the folliwing I characterize these transitions. This subsection documents the patterns of worker mobility across segments that motivate the spatial and occupational structure of the model. The facts establish that mobility differs across occupations; that worker flows decline with both geographic and occupational distance; that the

Table 3: Mobility Table

EmpTrans	MoveBoth	MoveLoc	MoveOcc	Stay
EE	4.0	12.1	21.0	62.9
NO	0.7	2.2	18.8	78.3
UE	6.5	14.4	35.7	43.4

moves workers actually make connect segments with nearly identical unemployment rates; and that movers require compensation in the form of higher wages. Appendix **??** shows that the sorting of destinations is also shaped by the occupational composition of destinations, conditional on individual characteristics.

Worker flows are shaped by both geographic and occupational considerations. I estimate a gravity model between all 148 location-occupation segments, aggregating worker transitions into flows between origin-destination pairs indexed by j and k. In the full specification, I include indicator variables for changing occupation and changing location, along with distances in both dimensions:

$$\log F_{jk} = \alpha \mathbf{1}\{\text{Change Occupation}\}_{jk} + \beta \mathbf{1}\{\text{Change Location}\}_{jk} + \gamma_1 \ln(\text{OccDist}_{jk}) + \gamma_2 \ln(\text{Dist}_{jk}) + \mu_j + \nu_k + \varepsilon_{jk}. \quad (3)$$

I estimate equation (3) in two ways. The first is OLS on the log of flows, the standard gravity specification, which conditions on the origin–destination pairs with positive flows. The second is Poisson pseudo-maximum likelihood (PPML), which estimates the same specification for the level of flows, $E[F_{jk}] = \exp(\alpha \mathbf{1}\{\text{Change Occupation}\}_{jk} + \cdots + \mu_j + \nu_k)$, and therefore retains the zero-flow pairs that the logarithm discards. Because zero flows are not random — pairs that are far apart in both dimensions rarely exchange any workers — the OLS estimates, which condition on positive flows, are potentially distorted by selection, whereas PPML remains consistent in the presence of heteroskedasticity.

Table 4 reports the estimates for EE and UE flows. Panel A shows the OLS estimates. Worker flows decline with both geographic and occupational distance: in the specification with geographic distance only, a one-unit increase in log distance lowers EE flows by 20% and UE flows by 17%. Adding occupational distance raises the R^2 from 0.47 to 0.74 for EE flows and from 0.45 to 0.72 for UE flows, confirming that both the geographic and the occupational dimensions matter for worker mobility. Adding the change indicators raises the R^2 further, to 0.78 for EE and 0.75 for UE flows. Conditional on the continuous distances, transitions that change location or occupation carry a large positive shift of 3–7 log points, and the distance coefficients steepen: the continuous distances capture the decay in flows among movers, while the change indicators capture the discrete level difference between transitions within and across location and occupation boundaries.

Panel B shows the PPML estimates on the level of flows, which retain the zero-flow pairs that the logarithm discards. The PPML distance elasticities are larger in magnitude than the OLS elasticities—for example, −0.82 versus −0.79 for EE geographic distance—and including segment fixed effects steepens the decay further. The PPML results therefore confirm the gravity pattern and show that worker mobility differs systematically across segments.

Table 4: Gravity regressions for EE and UE worker flows between the 148 segments. Panel A: OLS on log flows, conditioning on positive-flow pairs. Panel B: Poisson pseudo-maximum likelihood (PPML) on the level of flows, which retains zero-flow pairs and is robust to heteroskedasticity. OLS models include origin and destination segment fixed effects and progressively add occupational distance and change indicators; PPML is estimated without and with segment fixed effects.

	Panel A: OLS ($\ln F_{jk}$)						Panel B: PPML (F_{jk})			
	EE flows			UE flows			EE flows		UE flows	
Coefficient	Geo	+Occ	+Int	Geo	+Occ	+Int	no FE	FE	no FE	FE
ln(Geo. dist.)	−0.200***	−0.208***	−0.794***	−0.167***	−0.181***	−0.674***	−0.272***	−0.824***	−0.251***	−0.865***
ln(Occ. dist.)	—	−0.463***	−0.912***	—	−0.375***	−0.697***	−0.420***	−0.723***	−0.340***	−0.626***
Change occupation	—	—	5.232***	—	—	3.687***	—	3.269***	—	3.120***
Change location	—	—	6.325***	—	—	5.300***	—	5.882***	—	6.553***
R^2 (OLS)	0.474	0.740	0.778	0.445	0.715	0.746				

Geo = geographic distance only; +Occ = adds occupational distance; +Int = adds change indicators. OLS includes origin and destination segment fixed effects; PPML: no FE / FE = without / with segment fixed effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

To assess how much of the variation in transition hazards is associated with observable individual characteristics rather than with the segment in which a worker is employed, I also estimate Poisson hazard regressions for EE and UE moves with individual characteristics and origin-segment fixed effects. Table 5 reports the results. Individual characteristics matter—age lowers both hazards and education raises both—but their effects are small relative to the dispersion of the segment fixed effects: the range of the segment fixed effects is 4.6 times the largest individual effect for EE moves and 3.7 times for UE moves. Systematic differences across labor market segments, rather than the composition of workers, are the dominant source of variation in mobility.

3 Model

The model that I use to interpret these patterns of mobility across labor market segments is a wage-posting model with a segmented labor market. Workers search both when employed and when unemployed. Job offers are generated by poisson processes in each labor market segment. An offer is characterized by a triplet of location, occupation, and wage. Workers are forward-looking when evaluating the value of job offers. This means that they consider both the flow utility from wage and non-wage amenities, while also accounting for how accepting the offer is expected to affect the offers they will receive in the future.

Table 5: Poisson hazard regressions for EE and UE transitions. Panel A: coefficients on individual characteristics. Panel B: dispersion of the 147 origin-segment fixed effects relative to the largest individual effect. The segment fixed effects capture systematic variation in transition hazards across labor market segments after conditioning on individual characteristics.

Panel A: Individual characteristics		
Coefficient	EE hazard	UE hazard
Age (per year)	-0.012^{***}	-0.009^{***}
Experience (per year)	-0.019^{***}	0.014^{***}
Homeowner	-0.036^{***}	0.068^{***}
Female	-0.120^{***}	-0.060^{***}
Education (ISCED 6)	0.113^{***}	0.087^{***}
Panel B: Segment fixed effects (147 origin segments)		
Std. dev.	0.354	0.123
Range (max − min)	1.806	0.643
Max |individual effect|	0.397	0.172
Seg. range / max indiv.	**4.6**×	**3.7**×

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

When worker are hit by a separation shock, they enter a notice state, during which they continue to receive offer. As for the employed worker, the rate at which offers arrive depends on where the worker is located in location-occupational space, which implies that the model can generate escape margin that differ across segments.

The model is intended to highlight the nature of workers' job search and transitions along the two dimensions. It abstracts from general equilibrium mechanisms on the housing market that could be affected by workers' migration decisions and also does not cover the strategic interaction between workers' job search and firms' wage posting behavior.

The model is set in continuous time and the analysis focuses on the steady-state of the economy. The steady-state assumption is supported by the descriptive evidence in section 2 showing that the cross-sectional variation in unemployment rates are highly persistent over time.

Labor market segments The labor market consists of L geographic locations indexed $l \in \mathcal{L}$ and O occupations indexed by $o \in \mathcal{O}$. Each pair, (j, o), defines a distinct segment of the labor market, so that the economy can be viewed as a collection of $J \times O$ submarkets. Within a segment, $s \in \mathcal{S}$, jobs are characterized by a wage w drawn from a segment-specific distribution $F_s(w)$ with common support $[\underline{w}, \overline{w}]$.

Workers Workers can be either employed (i=e) or unemployed (i=u). Employed workers are located in the labor market segment of their workplace, while I assume that unemployed workers stay in the segment of their most recent workplace. This implies that neither employed nor unemployed workers can move to a different segment without first receiving and accepting a job offer in that segment.

The per-period utility of an employed worker in segment s is given by

$$u_s^e(w) = \gamma_{l(s)} + \eta_{o(s)} + \beta w, \tag{4}$$

which consist of location-specific amenities, γ_j, the nonpecuniary utility of working in occupation o, η_o^e, and the pecuniary utility of wage, βw. $l(s)$ and $o(s)$ are the location and occupation of segment s.

The economic rationale for including location and occupation amenities, denoted by γ_j and η_o, is that they capture the average utility associated with a particular location and occupation. These parameters absorb persistent utility components that are not otherwise represented in the model through wages, expected future job opportunities, or separation risk. Location amenities may reflect housing costs, access to public services, commuting conditions, and the availability of consumption and cultural amenities. Occupational amenities may capture working conditions, hours and flexibility, and physical demand. In the estimation, these amenity parameters allow the model to match persistent differences in employment shares and aggregate flows into and out of locations and occupations.

The flow utility of being unemployed in segment s is given by

$$u_s^u = \gamma_{l(s)} + \eta_{o(s)} + b, \tag{5}$$

where b is the flow utility of unemployment benefits.

Job offers and mobility costs Job offers arrive according to a Poisson process. The arrival rate is controlled by three parameters: the segment-specific arrival rate, $\lambda_{s'}^i$, that captures the job creation intensity of the destination segment (j', o'). That is, $\lambda_{s'}^i$ measures how frequently new job openings arise in segment (s'), independent of where workers are currently located.

A worker in segment s do not observe all such openings, but only a fraction that depends on the spatial and occupational frictions between the worker's current segment s and the destination s'. Let $\tau_{l,l'}^i$ denote the spatial accessibility between locations l and l', and $\omega_{o,o'}^i$ the occupational accessibility between occupations o and o'.

Combining these three parameters give the effective arrival rate of offers from destination segment s' for a worker currently employed in s by

$$\Lambda_{s,s'}^e = \tau_{l(s),l(s')}^e \, \omega_{o(s),o(s')}^e \, \lambda_{s'}^e. \tag{6}$$

Analogously, for unemployed workers searching for a job from a segment (j, o), where they were most recently employed,

$$\Lambda_{(s,s')}^u = \tau_{l(s),l(s')}^u \, \omega_{o(s),o(s')}^u \, \lambda_{s'}^u. \tag{7}$$

The effective arrival rates (6 and 7) parallels a gravity pattern of worker flow, in which observed mobility depends on both the mass (job opportunities) of the destination and the distance (frictions) between origin and destination. The accessibility parameters are allowed to vary between employed and unemployed workers which is a possible explanation for the differences in mobility across the two groups documented by the gravity equations.

Workers who have received a separation shock search while still employed. The effective arrival rate of offers to a terminated worker in segment s is

$$\Lambda^{n}_{s,s'} = \psi\big[\chi\,\Lambda^{e}_{s,s'} + (1-\chi)\,\Lambda^{u}_{s,s'}\big], \tag{8}$$

where $\psi \geq 1$ scales arrival rates to capture the potential increase in search effort after the shock, and $\chi \in [0,1]$ determines whether the destination composition of these offers follows the employed ($\chi = 1$) or the unemployed ($\chi = 0$) search structure. The baseline sets $\chi = 1$: terminated workers search from employment, so their opportunities arrive with the same structure — from the same network of firms and destinations — as ordinary on-the-job offers, only more frequently.

This choice is supported by evidence on how offer quality varies with search duration. Arcidiacono et al. (2023), estimating continuous-time search models with separate wage offer distributions on and off the job, find that longer unemployment durations are associated with substantially worse wage offers and lower offer arrival rates, so that the opportunities of unemployed workers deteriorate as the spell extends. Workers in the notice state search from employment, before this deterioration sets in, which supports tying their arrival structure to the employed one ($\chi = 1$).

A key feature of the model is that the arrival rates of offers depend on the network of occupations and locations such that workers must be strategic when deciding whether to accept or reject an offer since it has implications for the which job offers they will receive in the future. In regard to the notice state, this has a particularly nice interpretation since it implies that being in a dense labor market segment with a high arrival rate, acts as an insurance against unemployment. This mirror the empirical finding in Moretti and Yi (2024), that shows that workers in dense labor market recover faster after an separation shock.

Upon accepting an offer from segment s', the worker incurs a one-time mobility cost, $\kappa^{i}_{s,s'}$, which is an increasing function of the distance between segment s and s' in both the geographic and occupational dimensions. The mobility cost captures the direct costs of moving, such as relocation expenses, as well as the indirect costs associated with changing jobs, such as the loss of occupational-specific human capital or the need to adapt to a new work environment.

The parameterization of search frictions and moving costs is discussed in section 4.

Value functions The lifetime expected utility for an employed worker in segment s is represented by the Hamilton-Jacobi-Bellman (HJB) equation:

$$\rho V_s^e(w) = u_s^e(w) + \sum_{s' \in \mathcal{S}} \tau_{l(s),l(s')}\, \omega_{o(s),o(s')}\, \lambda_{s'} \int \mathbb{E}\big[\max\big\{V_s^e(w) + \varepsilon_r,\ V_{s'}^e(w') - c_{s,s'}^e + \varepsilon_a\big\}\big] - V_s^e(w) dF_{s'}(w') + \delta_{j,o}\big(V_s^n(w) - V_s^e(w)\big) \tag{9}$$

which is the expected lifetime utility of being employed in a given segment s with a wage w, which consists of the current utility flow, the expected value from receiving job offers, and the risk of receiving a separation shock, which moves the worker into a notice state described below rather than directly into unemployment. The expected value of job offers depends on how frequently they occur which depends on the location of the worker as given by (6), the distribution of offered wages, and the moving costs ($c_{s,s'}^e$).

The value of being unemployed follows the same structure as (10), but without the risk of receiving a separation shock. The expected lifetime utility of being unemployed in segment s is given by:

$$\rho V_s^u = u_s^u + \sum_{s' \in \mathcal{S}} \tau_{j(s),j(s')}^u\, \omega_{o(s),o(s')}^u\, \lambda_{s'}^u \int \mathbb{E}\big[\max\big\{V_s^u + \varepsilon_r,\ V_{s'}^e(w') - \kappa_{s,s'}^u + \varepsilon_a\big\}\big]\, dF_{s'}(w') \tag{10}$$

The preference shocks associated to the two options of rejecting or accepting, ε_r and ε_a, are drawn from the extreme value type-I distribution with scale parameter of 1.

The notice state has its own value function, $V_s^n(w)$. A worker in segment s with wage w who has received a separation shock keeps the flow utility of employment, receives offers at the rate determined by (8), and moves into unemployment at rate σ when the match ends:

$$\rho V_s^n(w) = u_s^e(w) + \sum_{s' \in \mathcal{S}} \Lambda_{s,s'}^n \int \mathbb{E}\big[\max\big\{V_s^n(w) + \varepsilon_r,\ V_{s'}^e(w') - c_{s,s'}^e + \varepsilon_a\big\}\big] - V_s^n(w) dF_{s'}(w') + \sigma\big(V_s^u - V_s^n(w)\big) \tag{11}$$

The notice period is an institutional feature of the Danish labor market: under the Salaried Employees Act, workers are entitled to between one and five months of notice depending on tenure, during which they search while still employed. In the baseline estimation, I set $\sigma = 4$, which corresponds to a three-month notice period which match that workers in the data have an average tenure within their employment spell of 3.5 years, which entitles them to three months of notice.

The notice state changes workers' acceptance behavior in two ways. First, the value of remaining in a terminated match is affected negatively by the imminent risk of entering unemployment: an offer is accepted when it beats $V_s^n(w)$,

which lies below $V_s^e(w)$. Terminated workers therefore accept offers they would have declined from ordinary employment, and wage cuts become acceptable – though the reservation value retains the flow utility of the current wage, so high-wage workers in the notice state remain more selective than low-wage workers. Second, offers arrive at the scaled rate $\psi \geq 1$: during notice, workers search can search more intensely. The probability of escaping rises through both margins, and the two are distinguishable. ψ governs the volume of escapes, while the lower acceptance bar governs their wage changes and destination composition. Entry into the notice state is itself not a choice: it arrives as a shock. What the data records as a separation into unemployment is therefore a shock together with a failed search, and the search was conducted with the urgency described above.

Endogenous search and wage setting Search in the model is endogenous on two margins. On the intensive margin, a worker in the notice state searches more intensely than an ordinary employed worker: the arrival rate of offers is scaled by $\psi \geq 1$ (equation 8), estimated at 17.2, so that notified workers search roughly 17 times more intensely than employed workers. On the extensive margin, the acceptance decision is endogenous: because the value of remaining in the terminated match lies below the value of ordinary employment, the reservation wage falls when the worker enters the notice state, and offers that would be rejected from employment are accepted from notice.

The model abstracts from wage bargaining. As in Burdett and Mortensen (1998), firms post wages and workers take them as given, with offered wages drawn from the segment-specific distribution $F_s(w)$. Entering the notice state affects the wages a worker faces through three channels. First, during the notice period the worker continues to earn her current wage: the flow utility $u_s^e(w)$ is retained, so the income from staying put is unchanged and the only new risk is the notice end rate σ. Second, the reservation wage falls, so a terminated worker accepts wage cuts that she would have rejected while employed; the escape margin therefore operates partly through lower accepted wages. Third, if the worker escapes into a new job, her wage is redrawn from the destination segment's distribution $F_{s'}(w')$, so the expected wage change depends on the wage distribution of the destination segment. Because the notice state retains the flow utility of the current wage, high-wage workers remain more selective than low-wage workers, and the wage cuts accepted in the notice state are concentrated among low-wage workers.

Had wages been set by bargaining instead of posted, the notice state would additionally shift the worker's outside option and thus the surplus split. Cederlöf et al. (2025) show that, under Coasian bargaining, firms substitute severance pay for advance notice, so that the wage effects of notice operate through the separation payment rather than through search while employed. Abstracting from bargaining therefore isolates the search margin, which is the focus of this paper.

The notice state and reallocation shocks The notice state should be distinguished from the reallocation shocks studied in the job-ladder and reallocation literatures. A reallocation shock—a "godfather shock," in the terminology of Autor et al. (2023)—is an exogenous event that hits an employed worker and moves her to a new job with a freshly drawn wage, "a job offer you cannot refuse," with unemployment as the alternative. In their model, separations decompose as $S(w) = \delta + \chi + \lambda^e(1 - F(w))$, where δ is the exogenous separation rate into non-employment, χ is the godfather-shock rate into another—possibly worse-paying—job, and the last term captures voluntary quits to better offers. The job-ladder models of Moscarini and Postel-Vinay (2018) rest on the same architecture: job destruction and reallocation across the ladder are two distinct processes, each with its own incidence. Because the EU and EE flows are driven by separate shocks, such models do not, on their own, generate a positive correlation between EU and EE transition rates: any comovement between the two flows has to be built into the joint incidence of the two processes.

The notice state in this paper is different. It is not a separate shock: it is a window that opens when the ordinary separation shock arrives, and whether the shock ends in EU or in EE is resolved endogenously by the worker's search during the window. The noticed worker makes the same decisions as employed and unemployed workers—she evaluates arriving offers against the same value functions, accepts or rejects them on the same terms, and searches with the same technology—but under different conditions. The imminent risk of unemployment, summarized by the notice end rate σ, lowers the value of remaining in the terminated match below the value of ordinary employment, $V_s^n(w) < V_s^e(w)$, so the reservation wage falls and search intensifies ($\psi \geq 1$).

This is what ties the escape margin to market conditions. The probability that a noticed worker in segment s makes an EE transition before the notice period ends depends on the effective arrival rate of offers, $\Lambda_{s,s'}^n$ (equation 8), which varies across segments with the intensity of job creation ($\lambda_{s'}^e$, $\lambda_{s'}^u$) and with the spatial and occupational frictions (τ, ω). Since the incidence of the underlying separation shock, δ_s, also varies across segments, both the EU rate, $\delta_s(1 - e_s)$, and the escape-driven component of EE flows, $\delta_s e_s$, comove with δ_s and with segment-specific market conditions. The model therefore generates the positive correlation between the observed EU and EE transition rates documented in Section 2 (Figure 4 and Table 2) as an endogenous outcome of a single shock process interacting with segment-specific labor market conditions, rather than as an exogenous correlation between two independent processes. This is what allows the model to separate the incidence of separation shocks from the escape margin, and to quantify how much of the cross-segment variation in unemployment is attributable to job-finding opportunities that are hidden inside the measured EU rate.

The advantage of formulating the model in continuous time is that I do not have to consider the scenario where more than one offer arrives within the same period. Instead each choice situation is binary and is related to non-negative expected value according to the HJB equations above. I further assume that the preference shocks are drawn from a extreme value type-1 distribution. This

implies that the discontinuous max operator can be replaced by a smooth logsum term, which expresses the expected value of the maximum of the two options (reject or accept) presented to the worker:

$$\mathbb{E}\left[\max\{V^e_{j,o}(w)+\varepsilon_r, V^e_{j',o'}(w')-c^e_{(j,o)\to(j',o')}+\varepsilon_a\}\right] \tag{12}$$

$$=\log\left(\exp\left(V^e_{j,o}(w)\right)+\exp\left(V^e_{j',o'}(w')-c^e_{(j,o)\to(j',o')}\right)\right), \tag{13}$$

which shows that the expected value of any offer is a continuous and strictly positive function of the offers' observable characteristics. The economic intuition behind this smoothing is that it captures the idea that workers may have some idiosyncratic preference for accepting an offer, and that even if an offer is not strictly better than the current job in terms of observable characteristics, it will still have a positive expected value due to the random preferences.

4 Results

This section presents the estimated model and uses it to quantify the mechanisms underlying cross-segment unemployment differences. I first report the estimated parameters and characterize the notice state. I then validate the model by comparing its steady-state implications with unemployment statistics in the data. Finally, I decompose the variation in unemployment rates across segments into the contributions of separation rates, job-finding from unemployment, and the escape margin, and I conduct counterfactual policy experiments.

4.1 Parameter Estimates

Table 6 reports the estimated parameters. The separation rate δ_s varies substantially across segments, with a mean of 0.092 per year (range: 0.024–0.260). This implies that the average employed worker in a segment experiences a separation shock roughly once every 11 years.

The offer arrival rate parameters λ^e_s and λ^u_s capture the intensity of job creation in each destination segment. The mean arrival rate for employed workers (0.188) is lower than for unemployed workers (0.503), consistent with the idea that unemployed workers search more intensely and therefore find more opportunities per unit of time. However, the arrival rates for unemployed workers are also more dispersed across segments (CV = 1.10 vs. 0.52), reflecting the fact that some segments are much better at absorbing unemployed workers than others.

The notice-state search intensity parameter ψ is estimated at 17.2. Since the arrival rate in the notice state is ψ times the employed arrival rate ($\chi = 1$), this implies that workers who receive a separation shock search roughly 17 times more intensely than ordinary employed workers. Combined with the notice end rate $\sigma = 4$ (corresponding to a three-month notice period, see Appendix ??), this generates an average escape hazard of 2.49 per year, or an average notice

duration of 1.85 months before either escaping to employment or transitioning to unemployment.

Table 6: Estimated Parameters

Parameter	Symbol	Estimate
Preferences		
Discount rate	ρ	0.040
Curvature (marginal utility of wage)	β	3.290
Unemployment utility	b	0.000
Separation and arrival rates (mean [min–max])		
Separation rate	δ_s	0.092 [0.024–0.260]
EE arrival scale (employed)	λ_s^e	0.188 [0.029–0.457]
UE arrival scale (unemployed)	λ_s^u	0.503 [0.007–3.165]
Notice state		
Search intensity	ψ	17.166
Notice end rate	σ	4.000
Destination mix	χ	1.000
Amenities (mean [min–max])		
Location amenity	γ_j	0.169 [0.033–0.233]
Occupation amenity	η_o	0.410 [0.004–0.914]
Search frictions (level, slope)		
Geographic, employed	τ_0^e, τ_1^e	2.089, 2.222
Geographic, unemployed	τ_0^u, τ_1^u	1.670, 2.114
Occupational, employed	ω_0^e, ω_1^e	3.140, 3.084
Occupational, unemployed	ω_0^u, ω_1^u	2.442, 2.304
Moving costs (level, slope)		
Geographic, employed	κ_0^e, κ_1^e	0.001, 0.002
Geographic, unemployed	κ_0^u, κ_1^u	0.332, 1.099
Occupational, employed	φ_0^e, φ_1^e	0.000, 0.001
Occupational, unemployed	φ_0^u, φ_1^u	0.002, 0.003

Notes: Estimates from structural estimation on DST data (2008–2019). Distributional parameters reported as mean across 148 segments, with range in brackets.

4.2 The Notice State and the Escape Margin

Table 7 summarizes the steady-state implications of the estimated model for the notice state. About 1.0% of workers are in the notice state at any point in time, with an average duration of 1.85 months. This is consistent with the statutory three-month notice period in Denmark ($\sigma = 4$ per year).

The key object of interest is the escape probability: the share of workers in the notice state who find a new job before their notice period ends. The

estimated model implies an escape probability of 38.4%, meaning that more than one in three separation shocks do not result in an unemployment spell. The remaining 61.6% of separations transition into unemployment.

This implies a structural EU rate of 0.041 per year, which is less than half the mean separation rate of 0.092. The difference is exactly the escape margin: $\delta_s \times e_s$ workers escape per year, while $\delta_s \times (1-e_s)$ transition into unemployment. The fact that the observed EU rate in the data (0.049 per year) is substantially below the separation rate is therefore consistent with a substantial escape margin.

Table 7: Notice State Statistics

Statistic	Value
Share of workers employed	0.914
Share of workers in notice	0.010
Share of workers unemployed	0.075
Average escape hazard (per year)	2.494
Notice end rate σ (per year)	4.000
Average notice duration (months)	1.85
Escape probability (notice $\rightarrow$ employment)	0.384
EU probability (notice $\rightarrow$ unemployment)	0.616
Escape EE share of separations	0.384
EU share of separations	0.616
Implied structural EU rate (per year)	0.041
Mean estimated separation rate δ (per year)	0.092

Notes: Steady-state distribution from the estimated model.

4.3 Model Validation: Fit to Unemployment Statistics

Table 8 compares the model-implied steady-state rates with the rates observed in the data across the 148 segments. The model fits the EU rate well (correlation of 0.97, mean of 0.056 vs. 0.049 in the data) and captures the EE rate reasonably well (correlation of 0.93). However, the model underpredicts the UE rate substantially (0.65 vs. 1.42 in the data, correlation of 0.72), which leads to an overprediction of the unemployment rate level (7.7% vs. 2.3% in the data, correlation of 0.64).

The model captures the cross-segment *pattern* of unemployment rates well but not the *level*. [There seems to be a data issue, because as shown unemployment rates are very stable in the this period, so I would expect the steady-state representation from the flows to match unemployment level, but 2% is too low]

Figure 5 shows the model-implied rates against the data rates for each of the four transition rates. The EU and EE rates lie close to the 45-degree line, confirming the good fit. The UE rate shows more scatter, with the model

Table 8: Model Validation: Segment-Level Rates (per year)

Rate	Data mean	Model mean	Data std	Model std	Corr.	RMSE
Unemployment rate	0.023	0.077	0.014	0.027	0.643	0.058
EU rate	0.049	0.056	0.034	0.034	0.973	0.011
EE rate	0.148	0.121	0.066	0.070	0.926	0.038
UE rate	1.422	0.651	0.400	0.247	0.722	0.820

Notes: Comparison of model-implied steady-state rates with data rates across 148 segments. Data rates computed from DST exposure (person-years) and event counts.

systematically below the data. The unemployment rate panel shows that the model captures the relative ordering of segments but overpredicts the level.

4.4 Variance Decomposition of Unemployment

To understand what drives cross-segment unemployment differences, I use the steady-state identity that holds almost exactly in the model ($R^2 = 0.9997$):

$$u_s = \frac{EU_s}{EU_s + UE_s}, \qquad EU_s \approx \delta_s(1 - e_s), \tag{14}$$

where u_s is the unemployment rate in segment s, EU_s is the flow rate into unemployment, and UE_s is the flow rate out of unemployment. The EU rate is approximately the separation rate times the probability of *not* escaping the notice state.

I decompose the variation in u_s across the 148 segments in two ways. First, I use the observable bathtub rates: the EU inflow rate and the UE outflow rate. Second, I decompose the EU inflow into its structural drivers with a Shapley decomposition of the R^2 from a regression of u_s on the separation rate δ_s, the escape probability e_s, and the UE rate UE_s, which attributes the explained variance in an order-independent way.

Table 9 and Figure 6 report the results. In the two-component decomposition, the EU inflow rate accounts for 73.9% of the explained variation in unemployment rates across segments, and the UE outflow rate accounts for 26.1%. In the three-component decomposition, the separation rate accounts for 71.0%, the UE rate for 22.5%, and the escape probability from the notice state for 6.5%. Job finding therefore operates through two margins—the standard UE channel and the escape margin—which together account for 29.0% of the explained variation, compared to 22.5% for the UE channel alone.

The explanatory power of the escape margin is modest relative to the separation rate, but it is not negligible. Although the escape probability varies less across segments than the separation rate, it captures a channel that is entirely hidden in the observable decomposition, where differences in job-finding opportunities are folded into the EU inflow rate. Since a large fraction of separations (38%) escape into employment, and Table 6 shows that EE arrival rates

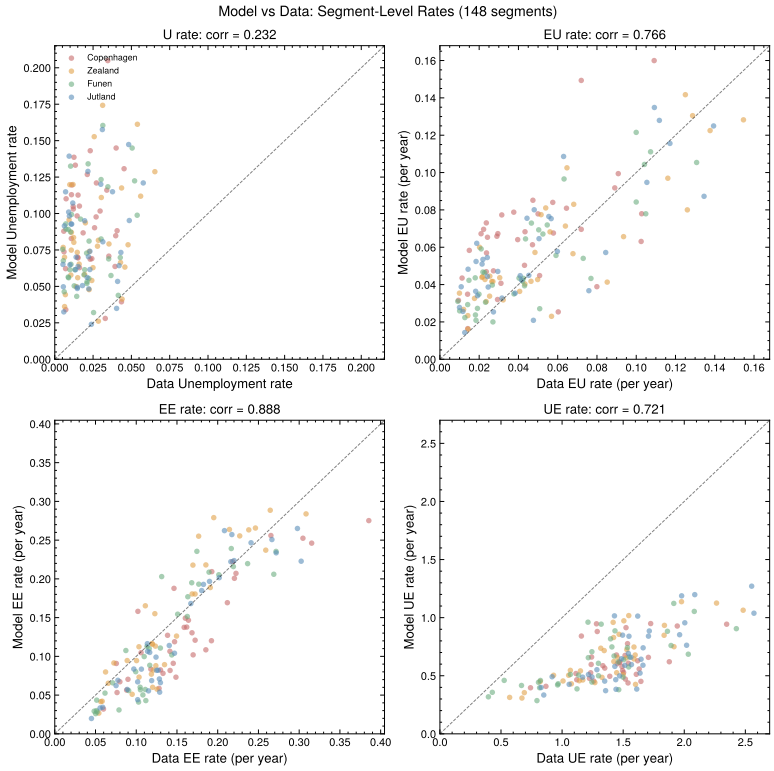


Figure 5: Model vs. Data: Segment-Level Rates. Each point is one of the 148 segments. Colors indicate the four model locations (Copenhagen, Zealand, Funen, Jutland). The dashed line is the 45-degree line.

Table 9: Variance decomposition of cross-segment unemployment rates. Three-component: Shapley decomposition of the R^2 from regressing segment unemployment rates on the separation rate, the escape probability from the notice state, and the UE job-finding rate (model $R^2 = 0.889$). Two-component: bathtub decomposition using the observable EU inflow and UE outflow rates (model $R^2 = 0.926$). The bathtub identity $u = EU/(EU+UE)$ holds with $R^2 = 0.9997$.

Component	Shapley R^2	Share of R^2 (%)	Model R^2
Three-component decomposition (structural)			
Separation rate (δ)	0.631	71.0	0.889
Escape from notice (e)	0.058	6.5	0.889
UE job-finding rate	0.200	22.5	0.889
Two-component decomposition (observable bathtub rates)			
EU inflow rate	0.684	73.9	0.926
UE outflow rate	0.242	26.1	0.926

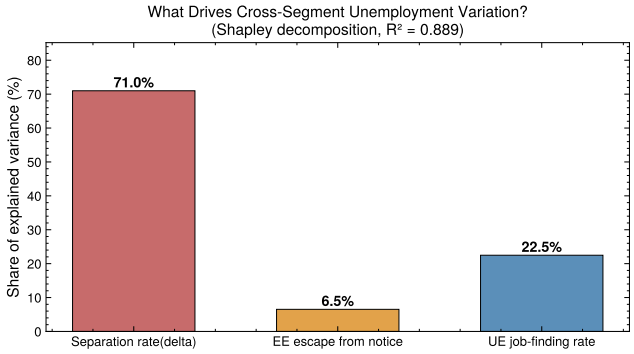


Figure 6: Shapley decomposition of the explained cross-segment variance in unemployment rates. Bars show the share of the model R^2 (0.889) attributed to the separation rate, the UE job-finding rate, and the escape probability from the notice state.

vary a lot across segments, even moderate cross-segment variation in escape probabilities translates into a visible contribution to unemployment differences.

5 Conclusion

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